Question Paper

Exam Date & Time: 30-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH END SEMESTER EXAMINATIONS, NOV/DEC 2024

DIGITAL COMPUTER ARCHITECTURE [ECE 3124]

Marks: 50 **Duration: 180 mins.**

DESCRIPTION

Answer all the questions.

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

1) Describe general register based computing machine using a neat block diagram showing the basic structure. Write 3- address instructions for computing the expression, (4)

A) $Y = (A - B) * (F * G + H) * (C - D)$ (assume memory operand is supported for source and destination).

B) Write (i) Load-Store based architecture and (ii) stack based architecture machine instructions for evaluating the following expression. (3)

$$Y = (A + B + C) * D * (E * (G - F)) + H$$

Given that the instruction *SUB src* will result in *Acc* *Acc - src* and *SUB* will result in (TOS-1 *minus* TOS) to be pushed on stack. Use Y as temporary storage wherever if required. Given that the maximum size of stack is two.

C) Assume that a portion of the direct mapped cache entry showing tags and line number looks like the table below. Identify each of the addresses as a cache hit or miss? (3)

a) 438EE816 b) F18EFF16 c) 6B8EF316 d) AD8EF316

Tag (binary)	Line number (binary)	Addresses wi/ block			
		00	01	10	11
0101 0011	1000 1110 1110 10				
1110 1101	1000 1110 1110 11				
1010 1101	1000 1110 1111 00				
0110 1011	1000 1110 1111 01				
1011 0101	1000 1110 1111 10				
1111 0001	1000 1110 1111 11				

2) Explain CLA adder along with the relevant logic expressions. Mention its advantages and disadvantages. (4)

A)

B) Write flowchart of Booth's algorithm for multiplying binary numbers. Multiply **-18** with **+8** showing all steps to find the product using the above algorithm. (3)

C) Construct a circuit using the required blocks for adding four 3-bit signed numbers using series-parallel hardware for CSA. Explain the working of the above circuit. How many clock pulses are required to get the final output of this circuit. (3)

3) Describe the 2 types of conditional I/O with relevant flowcharts. Why do we need interrupts in a computing system? (4)

A)

B) What is a TLB? Fill up the description row for the below mentioned table which has different events listed in each row. Write in the description, if such an event can occur or not and give a one line reason for your answer. (3)

TLB	Page Table	Cache	Description
Hit	Hit	Hit	
Miss	Miss	Miss	
Hit	Hit	Miss	
Miss	miss	hit	

C) Convert the decimal number 28.125 to 32-bit IEEE floating point representation. Write the final answer in hexadecimal. (3)

4) . Shown below are a sequence of instructions and a set of memory locations. Show the stack contents and SP contents after the execution of each instruction given below. Let the stack be empty initially and SP point to an address 75. Show the contents of memory locations 26, 27, 28 and 29 after the end of execution of pop B instruction. Assume SP increments for every push. (4)

A)

Address of the instruction	Instruction
12	Push A
14	Push B
16	Call Add_next
18	Pop A
20	Pop B

	Memory address	Data
	26	45
A →	27	96
B →	28	66
	29	88

B) Explain data hazard in pipeling with an example code and illustration of the pipeline showing the hazard and also showing the solution to overcome this hazard. (3)

C) For the address $6ABCDEA_{16}$, answer the following questions based on a 2-way set associative cache with 4K lines, each line containing 16 words, with the main memory of size 256 MB: (3)

a. What is the cache set number the identified block associated with the given address?

b. What will the tag be for the given address?

c. What will be the starting and ending address of the block in which the given address is

located?

5)		Explain Daisy chained method of interrupt polling using suitable diagrams. Explain its advantages and disadvantages.	(4)
	A)		
	B)	Explain and compare RISC and CISC ways of instruction set design.	(3)
	C)	Explain the concept of direct memory access with the help of a neat diagram. Explain the need for DMA in a computing system.	(3)

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